

KS3 Forces
Unit Overview

About This Unit
This unit aims to introduce students to forces by including hands-on investigations in each lesson. There is a focus on evaluating the investigations throughout the unit. Initially, students are guided step-by-step through writing an evaluation, then scaffolding is gradually reduced in subsequent lessons. Students are supported to rearrange equations and there are several opportunities to practice calculations through the unit.

The **Forces Homework Grid** provides suggestions for homework that will complement the unit.

Lesson 8 is a revision lesson which will help students to recap key words and concepts by playing engaging mini-games. There is also an **Escape the Room** lesson which can be used as an additional revision lesson. The **Forces Revision Pack** contains a knowledge organiser, differentiated revision mats and some differentiated practice exam questions.

There are two assessment packs to allow you to choose the level that best suits your students. The **Higher Assessment Pack** contains a more challenging version of the assessment, while the **Foundation Assessment Pack** contains scaffolding for students who may require more support. Both assessment packs include evaluation sheets for students to identify their strongest and weakest areas, plus a series of feedforward sheets that students can complete in the following lesson to build on their areas for development.

Lessons	Teacher Note:
1. Contact and Non-Contact Forces **2. Measuring Forces** **3. Gravity, Mass and Weight** **4. Air Resistance** **5. Friction** **6. Reducing Drag** **7. Hooke's Law** **8. Revision Lesson**	We hope you find the information on our website and resources useful. The activities set out in this resource are potentially hazardous. The activities are not suitable for all children and adult supervision may be required for some of the activities. It is your responsibility to assess whether the children in your care are able to safely carry out the activities and whether the children require adult supervision. You are responsible for carrying out proper risk assessments on the activities and for ensuring that activities can be carried out safely. We are not responsible for the health and safety of your group or environment so, insofar as it is possible under the law, we cannot accept liability for any loss suffered by anyone undertaking the activity or activities referred to or described in this resource. It is also your responsibility to ensure that those participating in the activity are fit enough to do so and that you or the organisation you are organising for has the relevant insurance to carry out the physical activity. If you are unsure in any way, we recommend that you take guidance from a suitably qualified professional.

Lesson 1 **Contact and Non-Contact Forces**

What's Included?

Teaching Ideas
Lesson PowerPoint
Investigating Forces Station Task Cards
Investigating Forces Results Table HA
Investigating Forces Results Table LA
Forces Glossary
Contact and Non-Contact Forces Worksheet

Resources

retort stand and ring magnets
matches and matchbox
cans e.g. baked beans
tin opener
cork
trough of water
hairdryer
table tennis ball
mixture of crushed cornflakes and iron filings in a clear zip-lock bag
bar magnet
eggshells (halves, washed)
house brick
balloon (inflated and tied)
small 1cm squares of paper
duster cloth
mini-whiteboards and pens (optional)

Learning Objective

To identify contact and non-contact forces.

Main Activities

Investigating Forces: Students visit a carousel of investigations to explore the different actions of some forces. They record their observations as diagrams and identify the forces that are acting in each scenario.

Contact and Non-Contact Forces: Students classify the forces as contact or non-contact and write a description of each category of force.

Success Criteria

- To list some forces.
- To distinguish between contact and non-contact forces.
- To investigate how different forces act.

Differentiation

There is a lower ability results table included which includes a diagram of each investigation and provides sentence starters for the students.

National Curriculum Links

Pupils should be taught about:

- forces as pushes or pulls, arising from the interaction between two objects;
- forces: associated with deforming objects; stretching and squashing – springs; with rubbing and friction between surfaces, with pushing things out of the way; resistance to motion of air and water;
- non-contact forces: gravity forces acting at a distance on Earth and in space, forces between magnets, and forces due to static electricity.

Health and Safety

We recommend that you have taken the time to read through each of the station instructions and set up the stations and equipment in advance of your lesson. You may choose to omit certain stations depending on a personalised risk assessment of your group and the location where the activity is being carried out.

Matches should be used in a well-ventilated area away from flammable materials. The classroom should have fire safety equipment on hand in case of an emergency.

Once opened, cans will be sharp. Students should be reminded to handle with care.

It is sensible to check whether students have any allergies before using food in a classroom. Students with an egg allergy should avoid touching the eggshells.

Hairdryers get hot with use, so students should turn them off between uses.

Lesson 2 **Measuring Forces**

What's Included?	Resources
Teaching Ideas **Lesson PowerPoint** **Force Diagrams** **Force Arrows** **Measuring Forces Worksheet HA** **Measuring Forces Worksheet LA**	newton meters objects to measure
Learning Objective	**Main Activities**
To identify and measure forces.	**Force Diagrams:** Students work together to add force arrows to three scenarios, prompting discussions over the position and size of arrows. **Balanced and Unbalanced Forces:** Students are introduced to the definitions and identify situations as balanced or unbalanced. **Measuring Forces:** Students measure classroom objects using newton meters and draw force diagrams to describe the investigation.
Success Criteria	**Differentiation**
• To identify forces. • To use force arrows in diagrams. • To compare balanced and unbalanced forces.	There is a lower ability activity sheet that contains some lower level questions.

National Curriculum Links	Health and Safety
Pupils should be taught about: • forces measured in newtons; • using force arrows in diagrams; • balanced and unbalanced forces.	When using newton meters, students should hold objects away from their feet or equipment to prevent damage if heavy objects fall.

Lesson 3 **Gravity, Mass and Weight**

What's Included?	Resources
Teaching Ideas **Lesson PowerPoint** **Technician Notes** **Gravity, Mass and Weight Worksheet HA** **Gravity, Mass and Weight Worksheet LA** **Gravity Calculations Worksheet** **Mass vs Weight Card Sort**	containers filled to a given mass newton meters to measure 0-5N
Learning Objective	**Main Activities**
To investigate what happens to mass and weight on different planets.	**Gravity, Mass and Weight:** Students are introduced to gravity and the difference between mass and weight. **Measuring Weight:** Students investigate the weight of containers that are filled to represent the different planets. **Weight Calculations:** Some questions are provided to challenge students' mathematical problem-solving skills.
Success Criteria	**Differentiation**
• To accurately measure weight using a newton meter. • To calculate weight and gravitational field strength. • To explain the difference between mass and weight.	The lower ability worksheet breaks down the calculations to support students in rearranging equations.

National Curriculum Links	Health and Safety
Pupils should be taught about: • forces measured in newtons; • gravity forces acting at a distance on Earth and in space.	Students should hold newton meters away from their feet when measuring the weight of the containers to avoid injury if containers fall.

Lesson 4 **Air Resistance**

What's Included?	Resources
Teaching Ideas **Lesson PowerPoint** **Air Resistance Illustrations** **Investigating Parachutes Method Sheet** **Investigating Air Resistance Worksheet HA** **Investigating Air Resistance Worksheet MA** **Investigating Air Resistance Worksheet LA** **Science Data Table Checklist** **Rank the Conclusion Cards**	material to make parachutes from (e.g. bin liners, cloth, carrier bags) string paperclips stopwatches sticky tape glue scissors

Learning Objective	Main Activities
To investigate air resistance in parachutes.	**Air Resistance:** Students are introduced to air resistance as a drag force and add arrows to represent the force to force diagrams. **Investigating Air Resistance:** Students investigate the effect of the size of a parachute on the time taken for the parachute to fall. **Conclusion and Evaluation:** Students are provided with support to write a conclusion and evaluation, including explanations of some of the working scientifically key words.

Success Criteria	Differentiation
• To describe the effect of air resistance on an object. • To investigate how the size of a parachute affects the time taken for a paperclip to fall. • To evaluate an investigation.	The middle ability and lower ability worksheets are scaffolded to support students in completing their results table, conclusions and evaluations. A table checklist is provided to help students self-assess their data tables.

National Curriculum Links	Health and Safety
Pupils should be taught about: • forces associated with pushing things away; resistance to motion of air and water. Pupils should be taught to: • make and record observations and measurements using a range of methods for different investigations; • evaluate the reliability of methods and suggest possible improvements; • apply mathematical concepts and calculate results; • interpret observations and data, including identifying patterns and using observations, measurements and data to draw conclusions; • present reasoned explanations, including explaining data in relation to predictions and hypotheses.	Students should be made aware of safe places they can drop their parachutes from.

Lesson 5 **Friction**

What's Included?	Resources
Teaching Ideas **Lesson PowerPoint** **Friction Investigation Method 1 Instructions** **Friction Investigation Method 1 Worksheet HA** **Friction Investigation Method 1 Worksheet LA** **Friction Investigation Method 2 Instructions** **Friction Investigation Method 2 Worksheet HA** **Friction Investigation Method 2 Worksheet LA** **Evaluation Key Words Card Sort**	wooden blocks a variety of materials (e.g. carpet, sandpaper, bubble wrap, cloth, foil) **either** pulleys, string and slotted masses **or** newton meters
Learning Objective	**Main Activities**
To investigate the amount of friction caused by different surfaces.	**Friction:** Students are introduced to friction with some demonstrations and discuss why friction is useful. **Investigating Friction:** Students investigate which surface provides the largest frictional force. **Conclusion and Evaluation:** Students are provided with support to write a conclusion and evaluation, including explanations of some of the key working scientifically words.
Success Criteria	**Differentiation**
• To investigate the effect of friction on an object. • To investigate how friction varies on different surfaces. • To evaluate an investigation.	There are higher ability and lower ability versions of the worksheets. On the LA sheets students are provided with key words and structured responses to help their explanations. HA students are given less scaffolding.

National Curriculum Links	Health and Safety
Pupils should be taught about: • forces associated with rubbing and friction between surfaces. Pupils should be taught to: • make and record observations and measurements using a range of methods for different investigations; • evaluate the reliability of methods and suggest possible improvements; • apply mathematical concepts and calculate results; • interpret observations and data, including identifying patterns and using observations, measurements and data to draw conclusions; • present reasoned explanations, including explaining data in relation to predictions and hypotheses.	Students should make sure not that they do not stand underneath the masses so that there are no injuries if they fall. A tray of sand or similar can provide a good surface for the masses to land on to prevent any damage to the floor.

Lesson 6 **Reducing Drag**

What's Included?	Resources
Teaching Ideas **Lesson PowerPoint** **Reducing Drag Method Instructions** **Reducing Drag Worksheet HA** **Reducing Drag Worksheet LA** **Resistive Forces Worksheet HA** **Resistive Forces Worksheet MA** **Resistive Forces Worksheet LA**	wallpaper paste **or** glycerol modelling clay string rulers stopwatches large measuring cylinders dry-wipe markers balances
Learning Objective	**Main Activities**
To investigate how to reduce resistive forces.	**Reducing Resistive Forces:** Students are given examples of how we can reduce resistive forces and introduced to the idea of streamlined objects. **Reducing Drag Investigation:** Students investigate which shape of modelling dough moves through wallpaper paste or glycerol the fastest. **Conclusion and Evaluation:** Students are provided with support to write a conclusion and evaluation, including explanations of some of the key working scientifically words.
Success Criteria	**Differentiation**
• To describe methods for reducing drag. • To investigate how to reduce drag forces. • To evaluate an investigation.	There are higher ability and lower ability versions of the investigation worksheet and HA, MA and LA versions of the plenary worksheets. On the LA sheets students are provided with key words and structured responses to help their explanations. MA students are given reminders of some key words and question prompts. HA students are given less scaffolding.

National Curriculum Links	Health and Safety
Pupils should be taught about: • forces associated with pushing things away; resistance to motion of air and water. Pupils should be taught to: • make and record observations and measurements using a range of methods for different investigations; • evaluate the reliability of methods and suggest possible improvements; • apply mathematical concepts and calculate results; • interpret observations and data, including identifying patterns and using observations, measurements and data to draw conclusions; • present reasoned explanations, including explaining data in relation to predictions and hypotheses.	Wallpaper paste may cause irritation to skin for some people. Students should be advised to wash any skin that comes into contact with the paste. Students should wear safety goggles for this activity. You are advised to check the data sheet for any chemicals you use in this activity.

Lesson 7 **Hooke's Law**

What's Included?	Resources
Teaching Ideas **Lesson PowerPoint** **Hooke's Law Method Instructions** **Hooke's Law Investigation Worksheet HA** **Hooke's Law Investigation Worksheet LA** **Graph Checklist** **Evaluation Worksheet HA** **Evaluation Worksheet LA**	graph paper slotted masses mass hangers springs clamp stands with boss and clamp G-clamps rulers
Learning Objective	**Main Activities**
To investigate how the force added to a spring affects the extension of the spring.	**Forces and Changing Shape:** Students are shown how forces can change the shape of objects. **Investigating Hooke's Law:** Students investigate how the extension of the spring is affected by the force applied to the spring. **Proportionality:** Students are introduced to the term 'directly proportional' and how this applies to Hooke's Law. **Calculating the Spring Constant:** Students rearrange the equation to calculate the spring constant. **Evaluation:** Students use differentiated sheets to evaluate their investigation.
Success Criteria	**Differentiation**
• To describe how forces can affect the shape of objects. • To investigate the relationship between force and extension. • To evaluate the investigation.	There are higher ability and lower ability versions of both worksheets. On the LA sheets students are provided with key words and structured responses to help their explanations. HA students are given less scaffolding.

National Curriculum Links	Health and Safety
Pupils should be taught about: • forces associated with deforming objects; stretching and squashing – springs; • forces measured in newtons; • measurements of stretch or compression as force is changed; • force-extension linear relation; Hooke's law as a special case. Pupils should be taught to: • make and record observations and measurements using a range of methods for different investigations; • evaluate the reliability of methods and suggest possible improvements; • apply mathematical concepts and calculate results; • interpret observations and data, including identifying patterns and using observations, measurements and data to draw conclusions.	Ensure that clamp stands are secured to the bench with G-clamps to avoid injuries from falling equipment.

Lesson 8 **Revision Lesson**

What's Included?	Resources
Teaching Ideas **Lesson PowerPoint** **Anagram Table** **Wipe Out Grid** **Collective Memory Graph** **Back to Back Illustrations**	whiteboards whiteboard pens
Learning Objective	**Main Activities**
To revise what you have learnt about forces.	**Anagrams:** Students recap some key words from the unit. **Wipe Out:** Students identify true and false statements about forces. **Collective Memory:** Students reproduce a graph based on Hooke's Law and then answer questions using it. **Back to Back:** Students draw forces diagrams and identify the forces on them, **Quiz, Quiz, Trade:** Students write questions and answers on forces and use them to test their peers.
Success Criteria	**Differentiation**
• To identify contact and non-contact forces. • To perform calculations involving forces. • To interpret a graph to answer forces questions.	You may select the activities from this lesson that are best suited for the ability of your group.

National Curriculum Links	Health and Safety
Pupils should be taught about: • forces: associated with deforming objects; stretching and squashing – springs; with rubbing and friction between surfaces, with pushing things out of the way; resistance to motion of air and water; • forces measured in newtons; • force-extension linear relation; Hooke's law as a special case.	Bags should be stored safely to avoid trips and falls when students are moving around the classroom.